Ideation Phase – Define the Problem Statements

Date: 25 June 2026

Team ID: XXXXX

Project Name: Smart Lender

Maximum Marks: 2 Marks

Customer Problem Statement

Banks and financial institutions receive many loan applications every day. Evaluating each application manually is time-consuming, requires extensive document verification, and may lead to inconsistent decisions. This process increases operational costs, delays loan approvals, and may result in human errors.

Loan applicants also face long waiting periods before receiving a decision, which affects customer satisfaction and trust in the lending process.

Therefore, there is a need for an intelligent system that can analyse applicant information such as income, education, employment status, credit history, loan amount, and property area to predict loan approval quickly and accurately.

Target Customers

- Banks
- Financial Institutions
- Loan Officers
- Credit Analysts
- Customers applying for personal, education, or home loans

Customer Pain Points

- Lengthy loan approval process
- Manual verification of applications
- Risk of human error and bias
- Difficulty in identifying creditworthy applicants
- High chances of loan defaults
- Delayed responses to customers

Proposed Solution

Smart Lender is a Machine Learning-based loan approval prediction system that automates the evaluation of loan applications. Using a trained **Random Forest** model, the system analyzes applicant details and predicts whether a loan is likely to be approved or rejected.

The application provides:

- Faster loan processing
- Improved prediction accuracy
- Reduced manual effort
- Consistent and data-driven decision making
- Better customer experience through instant prediction results

Expected Benefits

- Faster and more efficient loan approval process
- Reduced operational costs
- Lower risk of loan defaults
- Improved decision consistency
- Enhanced customer satisfaction
- Easy-to-use web application for both banks and applicants